

Unit - III

Equivalence of FDs:

defⁿ

* when 2 sets of FDs (E & F) are equivalent, it means E^+ equals F^+ , ($E^+ = F^+$)

- i) Every FD in ' E ' can be figured out from ' F '
 - ii) Every FD in ' F ' can be figured out from ' E '
- } E covers ' F '
and
' F ' covers ' E '

* Covering:

When a set of FDs say ' F ' covers another set of FDs say ' E ', it means that every FD in ' E ' can be figured out / derived from ' F '. (or) every FD in ' E ' is included in F^+ (closure).

* How to check if F covers E ? ($F \supseteq E$)

- i) Take each FD $X \rightarrow Y$ that exists in ' E '.
- ii) Cal X^+ using rules in ' F '.
- iii) Check if Y 's attr is included in X^+ .
- iv) If it works for every single FD in ' E ', F covers ' E '.

* are E & F equivalent?

- i) F covers E ($F \supseteq E$)
- ii) E covers F ($E \supseteq F$)
- iii) If both are true, they are equivalent.

$$\text{Q. } F = \{ A \rightarrow B, B \rightarrow C \}$$
$$E = \{ A \rightarrow B, B \rightarrow C, A \rightarrow C \}$$

i) F covers E ? $\Rightarrow$ Closure of 1st FD $\Rightarrow A^+ = ABC$ (check attr to ' E ' if present)
of ' E ' i.e. (A^+)
next $\Rightarrow B^+ = BC$

ii) E covers F ? $\Rightarrow$ Closure of 1st FD $\Rightarrow A^+ = ABC$
of ' F ' i.e. (A^+)
 $\Rightarrow B^+ = BC$

iii) Since $F \supseteq E$ and $F \subseteq E$, $F \equiv E$ // equivalent //

Q. $X = \{ AB \rightarrow CD, B \rightarrow C, C \rightarrow D \}$
 $Y = \{ AB \rightarrow C, AB \rightarrow D, C \rightarrow D \}$

i) X covers Y ? $\Rightarrow AB^+ = ABCD$
 $\Rightarrow C^+ = CD$

iii) $X \neq Y$ //

ii) Y covers X ? $\Rightarrow AB^+ = ABCD$
 $\Rightarrow B^+ = B$
 $\Rightarrow C^+ = CD$

Minimal Set of FDs:

* Minimal set of FDs 'F' should satisfy:

- (i) every FD in 'F' must end with only one attr on its R.H.S.
- (ii) we can't replace any FD $X \rightarrow A$ with a simpler $Y \rightarrow A$ (where Y is a smaller subset of X) & still keep the set equivalent.
- (iii) we can't remove any FD from 'F' and still keep the set equivalent.

* Minimal set is basically $\Rightarrow$ standardized/canonical form. (condition 1 makes sure this property)
 $\Rightarrow$ no redundancies. (condition 2 & 3 makes sure this property)

* 'Minimal Cover' $\Rightarrow$ minimal set of dependencies that is equivalent to a given set 'E', but in a standardized form, without any redundancy. There is always 1 minimal cover. If many min covers qualify, we use an extra criteria to choose b/w.

Q Find Minimal Cover for the F.D: $\{A \rightarrow B, C \rightarrow B, D \rightarrow ABC, AC \rightarrow D\}$.

Step 1 $\Rightarrow$ RHS should have only 1 attr. Separate & compare
 $\{A \rightarrow B \quad C \rightarrow B \quad \underline{D \rightarrow A} \quad \underline{D \rightarrow B} \quad \underline{D \rightarrow C} \quad \underline{AC \rightarrow D}\}$

Step 2 $\Rightarrow$ remove redundant FDs.

$A^+ = A \quad C^+ = C \quad D^+ = DBC \quad \underline{D^+ = DBCA} \quad D^+ = DAB \quad AC^+ = ACB$
 $A \rightarrow B \quad C \rightarrow B \quad D \rightarrow A \quad \underline{D \rightarrow B} \quad D \rightarrow C \quad AC \rightarrow D \Rightarrow \{A \rightarrow B, C \rightarrow B, D \rightarrow A, D \rightarrow C, AC \rightarrow D\}$
(say, we remove $A \rightarrow B$, (we can remove it)
check if we can get A^+ from other dependencies in the FD, excluding $A \rightarrow B$)
(since we do not have $A \rightarrow B$, we cannot remove it, not redundant)

Step 3: LHS should have only one attr:

$A \rightarrow B \quad C \rightarrow B \quad D \rightarrow A \quad D \rightarrow C \quad \underline{AC \rightarrow D}$
 $\hookrightarrow$ how to make it single?

(If in $\underline{AC \rightarrow D}$, we remove 'A', & if you take C^+ and get A in C^+ , we can remove 'A')

$C^+ = C$ (check in the above set)
(no 'A') (we cannot remove 'A').

(now do vice-versa)

$A^+ = A$ (no 'c', we cannot remove 'c' as well).

$\therefore$ we cannot further reduce step ②.

* Alg to find Min Cover:

Step ①

$\Rightarrow$ Break down the FD such that RHS should have only 1 attr, no multiple attr.

Step ② $\rightarrow$ remove redundant FDs, while keeping the set equivalent.

Step ③: LHS should have only one attr, while keeping the set equivalent. (again do step ②)

Q) $E: \{B \rightarrow A, D \rightarrow A, AB \rightarrow D\}$, find Min Cover?

$\Rightarrow$ ① $\{B \rightarrow A, D \rightarrow A, AB \rightarrow D\}$

$\Rightarrow$ ② $\{B \rightarrow A, D \rightarrow A, AB \rightarrow D\}$
($\because$ calculating trans)
 $B^+ = BD, D^+ = D, AB^+ = AB$

$\Rightarrow$ ③ $\{B \rightarrow A, D \rightarrow A, AB \rightarrow D\}$

$AB \rightarrow D \Rightarrow B^+ = BA$ (can remove A)
 $\therefore B \rightarrow D$ can replace $AB \rightarrow D$

$\therefore \{B \rightarrow A, D \rightarrow A, B \rightarrow D\}$

Ans: $\{B \rightarrow D, D \rightarrow A\}$

④ Step ② repeat $\Rightarrow$ $\{B \rightarrow A, D \rightarrow A, B \rightarrow D\}$
 $B^+ = BDA, D^+ = D, B^+ = BA$

$\therefore$ Min Cover = $\{D \rightarrow A, B \rightarrow D\}$

* Alg to find key 'K' for a set of FDs.

Step ① Identify all attr in 'R' from the given FD. 'K' is the 'R' of the FD.

Step ② Try removing each attr one at a time; if the remaining attr still form a superkey, remove that attr.

Q) finding key in $\{B \rightarrow D, D \rightarrow A\}$ above ques:

Step ① $\Rightarrow R = \{A, B, D\}$

$K = \{A, B, D\}$

FCP $\Rightarrow$ remove A: $B^+ = B \cup A$ $D^+ = A \cup D$ (all attr's can be derived w/o A)
 remove B: $A^+ = A$ $D^+ = D \cup A$ (not all attr's w/o B)
 remove D: $A^+ = A$ $B^+ = B \cup A$ (all attr's can be derived w/o D)

$\therefore K = \{B\}$ $\because B$ is the key as its min set of attr's that can derive all other attr's

Normal forms based on P.K.s:

- * While designing rel's we need:
 - a set of func dependencies for each relⁿ.
 - a designated pri key.
 - Tests for normal forms to guide Normalization process

* Introduced by Codd, 1972

- * Goals of Normalization:
 - Minimize redundancy.
 - Minimize insertion, deletion, update anomalies.

* a rel's N.F is the highest N.F condition it satisfies.

* 2 properties are crucial while normalizing:

- Lossless Join Property: nonadditive join (prevents spurious tuple generation)
- dependency preservation property (ensures all FDs are represented).

* rel's can sometimes be left in lower forms like 2NF for performance reasons. This is called 'Denormalization'. Designers do not always normalize to highest N.F.

Definitions for normalization:

- * Superkey: used to uniquely identify each tuple, redundant attr's are present. ex: $\{SSN, name, age\}$
- * Key: minimal - superkey. No redund^t attr's are present. ex: $\{SSN\}$
- * Candidate key: when a relⁿ has multiple keys.
- * Primary key: one candidate key chosen to be the default key / main identifier.
- * Unique / secondary key: All other candidate keys not chosen as pri key.
- * If no C.K is known, we can use the entire relⁿ as default S.K.
- * Prime Attr's:
 - Any attr that is a part of any candidate key.
- * Non-prime attr:
 - Any attr not a part of any C-key.

First N.F:

- * Each attrb in a tuple must contain only single / atomic value, no multivalued attrb in table.
- * not allowed: set of vals, tuples of vals, rel^{ns} within rel^{ns}.

ex:

Rollno	Name	Course
1	Sai	C/C++
2	Harsh	Java
3	Onkar	C/DBMS

→ not in 1NF

- * We have 3 solns to get in 1st N.F:

i) Convert it into two separate rows / tuples.

- * Make a combination of pri keys.

- * drawback: creates redundancy.

Primary Key

Rollno	Name	Course
1	Sai	C
1	Sai	C++
2	Harsh	Java
3	Onkar	C
3	Onkar	DBMS

(pri key cannot be null as it's repeating).
(we take {Course, prikey}).

ex 3

(a) DEPARTMENT

Deptname	Deptnumber	Deptmgr	Locations

(b) DEPARTMENT

Deptname	Deptnumber	Deptmgr	Locations
Research	5	333445555	(Bellare, Sugarland, Houston)
Administration	4	987654321	(Stafford)
Headquarters	1	888665555	(Houston)

(c) DEPARTMENT

Deptname	Deptnumber	Deptmgr	Location
Research	5	333445555	Bellare
Research	5	333445555	Sugarland
Research	5	333445555	Houston
Administration	4	987654321	Stafford
Headquarters	1	888665555	Houston

ii) Convert it into 2 separate columns / attrbs:

- * drawbacks: creates Null value.

- * requires knowing maximum no. of values.

- * makes queries more complex.

Primary Key: Rollno

Rollno	Name	Course1	Course2
1	Sai	C	C++
2	Harsh	Java	Null
3	Onkar	C	DBMS

(roll no. can be a pri key)
(use 'null')

iii) Create a separate relation. * considered the Best soln.

(referenced table)

- * One Base Table and another table for courses (referencing table).

P.K. → F.K (ref)

Rollno	Name	Rollno	Course
1	Sai	1	C
2	Harsh	1	C++
3	Onkar	2	Java
		3	C
		3	DBMS

Base table
Primary Key: Rollno

Primary Key: Rollno
Foreign Key: Rollno

(since nulls are not unique)

ex 2

(a) EMP_PROJ

Emp	Proj	Start	End	Prntbr	Hours
San	Enam	Prntbr	Hours		

(b) EMP_PROJ

Emp	Proj	Start	End	Prntbr	Hours
San	Enam	Prntbr	Hours		

(c) EMP_PROJ01

Emp	Proj	Start	End	Prntbr	Hours
San	Enam	Prntbr	Hours		

(d) EMP_PROJ02

Emp	Proj	Start	End	Prntbr	Hours
San	Enam	Prntbr	Hours		

SS → FK

2nd N.F:

* Full functional dependency:

⇒ $X \rightarrow Y$ is 'full' if removing any attr from 'X' breaks the dependency.

* Partial dependency:

⇒ a FD $X \rightarrow Y$ is 'partial' if any attr from 'X' can be removed & the dependency still holds.

* Rules: (i) Table shd be in 1NF.

(ii) All the non-prime attrs shd be fully functionally dependent on C-key.

(iii) no-partial dependency.

* 2NF only comes to picture when prim-key has multiple attrs. If a prim-key has a single attr, it is already in 2NF.

* How to find C-keys in a Table? ⇔ Closure Method

Q) $R(ABCD), FD \{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$

1st FD

⇒ $A^+ = ABCD$ (all attrs in R are present in A^+ , A can determine all dependencies)
∴ $CK = \{A\}$.

2nd FD

⇒ $B^+ = BCD$ X

∴ Prime attr = $\{A\}$.

⇒ $C^+ = CD$ X

∴ Non-prime = $\{BCD\}$

⇒ $D^+ = D$ X

⇒ $(AB)^+ = ABCD$ ✓ (also can be C.K.)

↳ But C.K shd be minimal.

↳ cannot be a C.K.

↳ But it is a S-key.

Q) $FD = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$

⇒ $A^+ = ABCD$

∴ CKey = $\{A, B, C, D\}$ //

⇒ $B^+ = BCAD$

∴ Prime attr ⇒ $\{A, B, C, D\}$ //

⇒ $C^+ = CDAB$

(used in C-key)

⇒ $D^+ = DABC$

∴ Non-prime attr ⇒ $\{\emptyset\}$ //

(not in C-keys)

Simpler method?

Q) $R = (ABCDE)$

$FD = \{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$ ('E' not present on right side so take E^+ first)

⇒ $E^+ = EC$

$$\Rightarrow (AE)^+ = ABECDE \quad \therefore CK = \{AE\}$$

$\Rightarrow$ What: (Others?? $BE^+, CE^+, DE^+ \dots$ but $\Rightarrow$ Time consuming) use this method instead

$$CK = \{AE\} \text{ (check if A, E are present in R.H.S.)}$$

$$(superse with that attr)$$

$$DE^+ = DEABC$$

$$\hookrightarrow BE^+ = BECA \checkmark \text{ (if we replace B with A, same L.K will arise, STOP process)}$$

$$\therefore CK = \{AE, DE, BE\}$$

(or)

$$ABCE^+ = ABCDE \text{ (superkey)}$$

$$CK = \{E\}$$

$$Q) \{A \rightarrow B, D \rightarrow E\}, R(A, B, C, D, E)$$

$$\Rightarrow ABCDE^+ = \{A, B, C, D, E\}$$

$$\Rightarrow ACD^+ = \{A, B, C, D, E\}$$

$$\Rightarrow ACD^+ = \{A, C, D, B, E\}$$

(we cant discard as nothing on RHS)

* Check if prime attr are present in RHS of FD, if no, it is a C-key.

$$\therefore CK = \{A, C, D\} \text{ (not on RHS) (if not, find function find + of subsets)}$$

$$Q) \{A \rightarrow B, B \rightarrow C, C \rightarrow A, A \rightarrow C\}, R(A, B, C, D)$$

$$\Rightarrow ABCD^+ = \{A, B, C, D\} \text{ (S-key)}$$

$$AD^+ = \{A, B, C, D\} \text{ (S-key)}$$

$$(cannot discard as it is determined then?)$$

$$AD^+ = \{A, B, C, D\} \text{ (S-key)}$$

$\hookrightarrow$ is it C-key?? $\Rightarrow$ Prime attr: $\{A, D\} \times$ C-key $\Rightarrow$ now take subsets.

$$\{A\}^+ = \{A, B, C\} \times$$

$$AD$$

$$\{D\}^+ = \{D\} \times$$

$$\{CD\}$$

$$CK \Rightarrow \{AD, BD, CD\}$$

* How to check if a FD is in 2NF or not?

$$\text{ex 1) FD: } \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E\}$$

$$1NF$$

$$R = (A, B, C, D, E, F)$$

(i)

$\Rightarrow$ it is in 1NF, yes.

$$(ii) \Rightarrow (AF)^+ = AFBCDE \checkmark$$

$$\therefore C.K = \{A, F\}$$

$$(iii) \Rightarrow \text{prime attr} = A, F$$

$$\Rightarrow \text{non prime attr} = B, C, D, E$$

$$\Rightarrow \text{for 2NF form, (proper subset of C.K} \xrightarrow{\text{det}} \text{non prime attr)}$$

in $\{A, F\}$ subsets are $A \rightarrow$ det B (non prime) $\therefore$ Partial dep exists $\therefore$ not in 2NF

$\Rightarrow$ partial dependency

ex 9) $FD = \{AB \rightarrow CD, C \rightarrow A, D \rightarrow B\}$

$R(A, B, C, D)$

i) $\Rightarrow 1NF$

ii) $\Rightarrow$ Since all are in RNS: (use the other method)

iii) $\Rightarrow AB \rightarrow CD = \{A B C D\}$

$AB^+ = \{A B C D\}$

$\hookrightarrow CB^+ = \{C B A D\}$

$\hookrightarrow CD^+ = \{C D B A\}$

} normal method

$CK = \{AB, CB, CD\}$

$\Rightarrow P.Atk = A B C D$

$\Rightarrow$ non-prime = X (no non primes only) $\therefore 2NF$ from

PM:

$\Rightarrow$ prime attr are a part of C-key.

$\Rightarrow$ are $\{CustID, StoreID\}$

$\therefore$ Prime attr = $CustID$

$StoreID$

$\therefore$ non-prime = location.

Customer		
<u>CustomerID</u>	<u>StoreID</u>	Location
1	1	Delhi
1	3	Mumbai
2	1	Delhi
3	2	Banglore
4	3	Mumbai

Candidate Key: CustomerID StoreID
 Prime Attributes: CustomerID StoreID
 Non Prime: Location

$\Rightarrow$ location is a partial depen as it is dependent only on a part of the C-key.

$\therefore$ not in 2F

So, divide the table

$\Rightarrow$ Store ID uniquely identifies location. So we can create a sep table for it.

Store	
<u>StoreID</u>	<u>Location</u>
1	Delhi
2	Banglore
3	Mumbai

ate Key.

<u>CustomerID</u>	<u>StoreID</u>
1	1
1	3
2	1
3	2
4	3

So this is not a problem, so this table this in.

$\therefore 2NF$ from as location is directly derived from C-key itself & not a 'part' or 'subset' of C-key.

3NF (remove insert, delete, update anomalies completely).

Conditions:

i) $\Rightarrow$ Should be in 2NF.

$\Rightarrow$ If a non prime attr determines non-prime attr. This means there is transitive dependency (for every non-trivial F.D)

$\Rightarrow$ If there is Transitive dependency is present in the reln, is not in 3NF.

ii) $\Rightarrow$ we have to make sure, no Transitive dep is present for 3NF.

(or)

for each F.D, $\left. \begin{array}{l} \text{LHS is S-key / C-key} \\ \text{RHS is prime attr} \end{array} \right\}$ atleast one condition holds.

ex: $R(A, B, C, D)$

FD: $A \rightarrow B \quad B \rightarrow C \quad C \rightarrow D$

i $\Rightarrow$ C-key = $\{A\}$

ii $\Rightarrow$ prime attr = A

$\Rightarrow$ non prime = B, C, D.

iii * check FDS if NPA ~~is~~ NPA. If yes, not in 3NF.

$A \rightarrow B$ X

$B \rightarrow C$ $\checkmark$ $\therefore$ not in 3NF.

(00)

* check if LHS is S-key $\Rightarrow$ 3NF //

(00)

or
Cand key

* check if RHS has prime attr $\Rightarrow$ 3NF //

$\rightarrow$ even if 1 NPA is present with P.A in LHS, it is still considered as NPA

Any one

ex: $R(AB(CDEF))$

FD: $\{AB \rightarrow CDEF, BD \rightarrow F, A \rightarrow D\}$

i $\Rightarrow$ C-K = $\{A, B\}$

ii $\Rightarrow$ prime = $\{A, B\}$ non prime = $\{C, D, E, F\}$

iii $\Rightarrow$ $AB \rightarrow CDEF$ $\checkmark$ (if LHS is P.A, no need to check RHS)

3 conditions $A \rightarrow D$ $\checkmark$ (~~~~~)

$BD \rightarrow F$ X

$\underbrace{BD}_{\substack{\text{P.A. N.P.A.}}} \rightarrow F$ $\therefore$ not 3NF

(if RHS has one P.A & one NPA; it is completely considered as NPA on LHS).

BCNF

Trivial: $(A)B \rightarrow (A)$

non-trivial: $AB \rightarrow D$ X.

$\Rightarrow$ Every relⁿ in BCNF is in 3NF form.

conditions

$\Rightarrow$ i should be in 3NF.

ii if each non-trivial FD $X \rightarrow Y$, X must be a S-key / C-key.
(not present on BS)
 $AB \rightarrow D$

ex: $R(A, B, C)$

FD: $\{A \rightarrow B, B \rightarrow C, C \rightarrow A\}$

i $\Rightarrow$ C-K $\{A, B, C\}$

ii $\Rightarrow$ S-K
 $(A) \rightarrow B$ $\checkmark$
 $(B) \rightarrow C$ $\checkmark$
 $(C) \rightarrow A$ $\checkmark$
 $\therefore$ BCNF form //

Properties of relational decompositions

Q) Convert a relⁿ to 2NF form / decompose into 2NF form:

① $R(ABCD) F = \{A \rightarrow B, B \rightarrow C\}$

Step 1 $\Rightarrow$ normally check if it is in 2NF:

$$ABCD^+ = ABCD$$

$$ABD^+ = ABCD$$

$$AD^+ = ABCD$$

$$\hookrightarrow A^+ = ABC$$

$$\hookrightarrow D^+ = D$$

(to check if AD^+ is a CK, check if the proper subsets are SK, if they are then AD^+ is not CK).
(not SK, $\therefore$ It is a CK)

Only 1 CK is present as the P.A are not on the RHS of FDs)

$$\therefore CK = AD$$

$$P.A = \{A, D\}$$

$\therefore$ the 1st FD is violating 2NF rules as there is partial depⁿ

Step 2 $\Rightarrow$ find closure of L.H.S of FD creating a problem & make sub-relns:

$$A^+ = ABC$$

$$R_1 = \{A, B, C\} \quad R_2 = \{D\}$$

} we have R_1 with closure etc. of 'R'
 R_2 has remaining from 'R'.

But it is not a proper decomposition & it shd be dependency preserving & lossless. But there are no common attrb b/w R_1 & R_2 . So, it is Lossy decom.

Step 3 add a common attrb. The common attrb should be a SK of atleast one sub-reln, then it is said to be Lossless decom.

$$R_1 = \{A, B, C\}$$

$$R_2 = \{A, D\}$$

$\rightarrow A^+ = ABC$ (A is SK of R_1 as all can be derived)

$\therefore$ we have decomposed into R_1 & R_2 //

Step 4 find FD of R_1 & R_2 :

$$R_1 = \{A, B, C\}$$

$$A^+ = ABC \quad AB^+ = X \text{ no need to check}$$

$$B^+ = BC \quad AC^+ = X \text{ check}$$

$$C^+ = C \quad BC^+ = BC$$

$$R_2 = \{A, D\}$$

$$A^+ = ABC \text{ (not a part of } R_2)$$

$$D^+ = D$$

$$\therefore \{A \rightarrow BC, B \rightarrow C\} = F_1 //$$

$$\therefore \{\emptyset\} = F_2 //$$

(since only 2 attrb are present R_2 is already in BCNF.
L.H.S has SK in the F.D)

$\therefore CK = \{A\}$
 $\therefore P.A = \{A\}$ $N.P.A = \{\emptyset\}$
 (If C.K is single then there will be no chance of P.Dcp as there will not be any proper subsets of A. Hence, no need to check rns)
 check $\therefore 2NF \checkmark //$

Step 5 Check dependency preservation: $F_1 \cup F_2 = F$

$\therefore F_1 \cup F_2 = \{\emptyset\} \cup \{A \rightarrow BC, B \rightarrow C\}$
 $= \{A \rightarrow BC, B \rightarrow C\}$
 $\therefore$ exists in $F //$

(ii) $R(ABCD) F = \{A \rightarrow B, C \rightarrow D\}$

$\Rightarrow ABCD^+ = ABCD$

$A^+ = ABCD$

$\hookrightarrow A^+ = AB$
 $\hookrightarrow C^+ = CD$

$\therefore CK = AC$

$\therefore P.A = \{A, C\}$ $\therefore$ only 1 C.K //

$\Rightarrow$ Both $A \rightarrow B$ & $C \rightarrow D$ have Partial dep.

$\Rightarrow A^+ = AB$ $C^+ = CD$ $R_3 = \{A, C\}$ (no remaining attributes)
 $R_1 = \{A, B\}$ $R_2 = \{C, D\}$ (do this)

(all to lossless, S.K of sub relⁿ should be common. So, take A i.e S.K of R_1 , add A to R_3 . Now check in R_2 , C is S.K. So put C in R_3 .)

$\therefore$ now decom is lossless //

$\Rightarrow$ now find if it is dep preserving or not after finding FDS:

$R_1 = \{A, B\}$

$A^+ = AB$

$B^+ = B$

$R_2 = \{C, D\}$

$C^+ = CD$

$D^+ = D$

$R_3 = \{A, C\}$

$A^+ = AB$

$C^+ = CD$

By default all in BCNF. But, you can do it manually

$F_1 = \{A \rightarrow B\} //$

$F_2 = \{C \rightarrow D\} //$

$F_3 = \{\emptyset\}$

$C.K = A$
 $2NF$ (BCNF (LHS = SK))

$C.K = C$
 $2NF$ (BCNF (LHS = SK))

$\Rightarrow F_1 \cup F_2 \cup F_3 = F$

$\{A \rightarrow B\} \cup \{C \rightarrow D\} = F //$ $\therefore$ Dep pres satisfied //

Q) Convert 2NF to 3NF / decompose into 3NF:

(i) $R(ABCD) F \{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$

$\Rightarrow$ check for 3NF normally:

$$ABCD^+ = ABCD$$

$$A^+ = ABCD \text{ (no proper subsets) //}$$

$$\therefore CK = A$$

$$\therefore P.A = \{A\} \text{ (only 1 CK present) //}$$

$\Rightarrow B \rightarrow C, C \rightarrow D$ are creating prob due to transitive depⁿ.

$$\Rightarrow B^+ = BCD$$

$$R_1 = \{B, C, D\}$$

$$B^+ = BCD$$

$$C^+ = CD$$

$$D^+ = D$$

$$BC \rightarrow$$

$$BD \rightarrow$$

$$CD = CD$$

$$C^+ = CD$$

$$R_2 = \{C, D\}$$

$$C^+ = CD$$

$$D^+ = D$$

$$F_2 = \{C \rightarrow D\} //$$

$$R_3 = \{A, B\} \rightarrow \text{why not C?}$$

$$A^+ = ABC$$

$$B^+ = BC$$

$$F_3 = \{A \rightarrow B\} //$$

$\{R_2 \text{ \& } R_3 \text{ are in BCNF already?}$

$$F_1 = \{B \rightarrow CD, C \rightarrow D\} //$$

$$\boxed{C.K = B}$$

$\therefore B \rightarrow CD$ does not have transitive dep but $C \rightarrow D$ does have //

$\therefore$ decompose R_1 again //

$$\Rightarrow R_1 = \{B, C, D\}$$

$$F = \{B \rightarrow CD, C \rightarrow D\}$$

$$C^+ = CD$$

problem $\rightarrow$ T. Depⁿ

$$R_{11} = \{C, D\}$$

$$F_{11} = \{C \rightarrow D\} //$$

$$\boxed{C.K = C}$$

$\therefore$ no transitive

$\therefore$ 3NF & BCNF //

$$R_{12} = \{B, C\}$$

$$F_{12} = \{B \rightarrow C\} //$$

$$\boxed{C.K = B}$$

$\therefore$ No trans

$\therefore$ 3NF & BCNF //

common for lossless

$\therefore$ Totally we have $R_1, R_2, R_3, R_{11}, R_{12} \Rightarrow R_{12}(B, C), R_2(C, D), R_3(A, B)$
 $\downarrow$ discard $\downarrow$ discard as $\{C, D\}$ already in R_2
 [also satisfies lossless & D. prop^{ies} Properties]

Dependency preservation property:

Q) $R = (A B C D E) \quad F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$.

$R_1 = (A B C)$

$R_2 = (C D E)$

$\Rightarrow A^+ = A B C D \quad A \rightarrow B C$

$B^+ = B C D A \quad B \rightarrow C A$

$C^+ = C D A B \quad C \rightarrow A B$

$AB^+ = A B C D \quad \text{dup} \quad \text{duplicate F.D as A alone can derive C}$

$AC^+ = X \quad \text{dup} \quad \text{(trick: if A can derive all attr's)}$

$BC^+ = X \quad \text{dup} \quad \text{sub set, no need to check for others}$

$\therefore F_1 = \{A \rightarrow B C, B \rightarrow C A, C \rightarrow A B\}$

$\Rightarrow C^+ = C D A B \quad C \rightarrow D$

$D^+ = C D A B \quad D \rightarrow C$

$E^+ = E$

$CD^+ = X$

$CE^+ = C E A B D \quad \text{dup} \quad \text{(as C \& D alone can derive the same)}$

$DE^+ = D E A B C \quad \text{dup}$

$F_2 = \{C \rightarrow D, D \rightarrow C\}$

condition

$\Rightarrow [F_1 \cup F_2 \equiv F] \Rightarrow \text{dep}^n \text{ preserving}$

$\Rightarrow \text{find } G \supseteq F \text{ and } F \supseteq G \text{ (as least unit for equivalence)}$

$\Rightarrow G = \{A \rightarrow B C, B \rightarrow C A, C \rightarrow A B, C \rightarrow D, D \rightarrow C\}$
 $F = \{ \text{question} \}$
 find if $G \equiv F$

$\therefore \text{Yes, dep pres satisfied}$